

Mounika Reddy

Software Engineer

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SUMMARY

Results-oriented Software Engineer – AI/ML with 5+ years of experience delivering scalable, secure, and production-grade ML and cloud-native solutions. Specialized in LLM-powered chatbots, semantic search, and RAG pipelines using LangChain, FAISS and transformer models. Skilled in Python microservices, ReactJS frontends, and AWS deployments. Experienced in CI/CD automation, Kubernetes orchestration, and secure application delivery with OAuth2 and JWT. Strong foundation in object-oriented programming, software architecture, and design patterns, with hands-on experience building scalable Python microservices using FastAPI, Flask, Django, and SQLAlchemy.

Technical Skills

LLMs & GenAI: LangChain, OpenAI APIs, Prompt Engineering, LangGraph, RAG, Semantic Search
ML & Deployment: Scikit-learn, Hugging Face, Sentence Transformers, TensorFlow, Model Evaluation
Search & Retrieval: FAISS, Embedding Generation, Document Chunking, Contextual Retrieval
Cloud & MLOps: AWS (EKS, ECS, EC2, S3), Docker, Kubernetes, Jenkins, CI/CD Pipelines
Full-Stack: Python (FastAPI, Flask, Django), Java, ReactJS, TypeScript, Redux, REST APIs
Architecture & Backend Design: Object-Oriented Design (OOP), Design Patterns, RESTful Architecture, SQLAlchemy, Django ORM
Data & Security: SQL, PostgreSQL, Oracle, Kafka, OAuth2, JWT, Secure API Design
Testing: Alteryx Designer, Pytest, Postman, Selenium, Load Testing, Monitoring (Prometheus)
Dev Practices: Agile (Scrum, JIRA), Git, Code Reviews, Automation, Compliance-first Development

Work Experience

Software Engineer – II Oct 2024 - Present
CVS Health Remote

- Built ReactJS interfaces with REST APIs, improving internal tool performance and increasing user engagement by over 35% organization wide.
- Optimized PostgreSQL schemas and indexes to improve query performance, reducing data retrieval time by 25% in production environments.
- Developed Python microservices using FastAPI, reducing release cycles by 25% and supporting agile deployment across dynamic system architectures.
- Designed and automated data workflows using Alteryx for campaign and operational analytics, integrating data from multiple sources (APIs, databases, and CSV files) to streamline performance reporting.
- Built Alteryx workflows with embedded Python scripts to perform data wrangling, transformation, and post-campaign performance tracking, reducing manual processing time by 40%.
- Implemented data access layers using SQLAlchemy and PostgreSQL, optimizing ORM-based queries and ensuring clean separation between business logic and persistence layers.
- Created unit and integration tests using Pytest and Postman, reducing production bugs by 40% and improving code quality.
- Configured CI/CD pipelines with Jenkins and Azure DevOps, enabling blue-green deployments and reducing deployment issues by over 50%.
- Deployed containerized applications on AWS EKS with autoscaling and rolling updates, ensuring 99.9% uptime during peak usage periods.
- Containerized microservices and front-end applications using Docker, streamlining development workflows, and improving portability across environments.
- Orchestrated deployments using Kubernetes on AWS EKS, implementing autoscaling, rolling updates, and ensuring 99.9% uptime during peak usage periods.

Application Developer Apr 2023 - Oct 2024
State of Missouri Jefferson City, MO

- Developed React and Redux dashboards for state inspectors, boosting accessibility and accelerating field reporting speed by 40% overall.
- Built and maintained Alteryx workflows for automating data extraction, transformation, and validation across multiple state systems, improving data accuracy and reducing manual processing.
- Tuned PostgreSQL performance and optimized complex queries, reducing report generation time by 30% and improving backend data efficiency.
- Re-architected a legacy monolith into scalable Python microservices, enhancing maintainability and enabling faster, more development cycles.
- Integrated automated testing into CI workflows, cutting manual QA workload by 50% and improving confidence in production deployments.
- Automated CI/CD processes using Azure DevOps, reducing deployment errors and increasing release reliability across production environments.
- Migrated backend services to AWS EKS with autoscaling and load balancing, increasing infrastructure resilience and uptime by 35%.

Earlier Experience - State of Missouri

- Developed Java-based REST APIs using Spring Boot and Hibernate, supporting mission-critical reporting and data integration.
- Migrated legacy applications to microservices architecture using Java and Kafka.

- Created scheduled data ingestion services from EPA systems using Java 8, improving data pipeline reliability.
- Used Maven, Azure DevOps, and Git for builds, deployments, and version control.
- Actively participated in security audits, fixed vulnerabilities, and supported best practices in code safety.
- Practiced Agile/SCRUM methodologies using Azure Boards.

Software Engineer

Dec 2022 - Apr 2023

Thinklusive Inc

New Jersey

- Built a TypeScript-based frontend using React, improving user task efficiency by thirty percent on internal dashboards.
- Refined Oracle database queries and introduced caching layers, successfully reducing backend response latency by thirty-five percent overall.
- Implemented Django REST APIs alongside Celery for asynchronous processing, increasing data throughput efficiency by thirty percent.
- Developed robust test coverage with Pytest and Selenium, maintaining full code coverage and ensuring deployment stability always.
- Containerized services with Docker and created CI/CD pipelines using AWS Code Pipeline, improving release velocity by forty percent.
- Hosted services on AWS ECS and managed storage assets in S3, ensuring scalability and reliable uptime for applications.

Programmer Analyst

Jan 2022 - Dec 2022

Ozarks Public Health Institute (OPHI)

Springfield, MO

- Created responsive frontend interfaces using JavaScript and Bootstrap, improving user experience and reducing data entry errors by 20%.
- Designed and managed SQL database schemas to streamline backend data handling, enhancing analytics accuracy and reporting efficiency.
- Used Alteryx for data preparation and transformation of health datasets, automating validation and improving report turnaround times.
- Built RESTful APIs with Flask to support health portal features, reducing request latency and improving overall system responsiveness by 25%.
- Tested API endpoints using Pytest and Postman, ensuring reliable functionality and contributing to consistent high-quality production releases.

Subject Matter Expert

Aug 2019 - Jul 2021

Amazon

Hyderabad, India

- Provided interface-level feedback on internal tools, leading to design improvements and a 15% boost in support agent efficiency.
- Verified data consistency across internal systems, enhancing operational trust and improving accuracy during customer support interactions.
- Collaborated on backend testing for translation tools, reducing user issues and improving system reliability across multilingual platforms.
- Delivered tech support and diagnosed system faults, decreasing ticket resolution times by 35% through proactive issue identification.
- Participated in release validation, increasing deployment reliability and reducing onboarding time for agents using internal support tools.
- Maintained AWS-hosted internal platforms, optimizing uptime and improving performance for customer-facing applications and internal support systems.

EDUCATION

Master of Science Computer Science

Aug 2021 - Dec 2022

Missouri State University

Springfield, Missouri

- Specialization: Data Science
- Coursework: Advanced Algorithms, Data Mining, Machine Learning, Advanced Databases, Deep Learning.

Bachelor of Technology Computer Science

Aug 2015 - May 2019

JNTUH

Hyderabad, India

- Coursework: Data Structures & Algorithms, Database Management, Cloud Computing, Big Data Analytics.

CERTIFICATIONS

- Data Science Certification.
- PMP® – Project Management Professional (PMI)
- Awarded Graduate Assistantship (2022) – Selected based on academic excellence and technical capability